

PHASE 1 PROJECT PROPOSAL & SRS

Web-Based Document Upload & Direct Printing Platform

Project Phase: Phase 1 – Core Printing Platform

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Target Delivery: Approximately 4 Weeks

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Document Control

Item	Details
Document	Phase 1 Project Proposal, Scope, SRS & Technical Architecture
Version	1.0
Status	Client Review / Proposal
Phase	Phase 1
Target duration	Approximately 4 weeks
Maintenance	1 year, as defined in Section 25
Operational costs	Excluded; payable separately by client

1. Executive Summary

This proposal defines the Phase 1 scope for a multi-shop web platform for stationery and printing shops. Customers will scan a shop-specific QR code, upload a document, select printing options, see a calculated price, and track the print status. The shopkeeper will receive the print request in a shop-isolated dashboard and can initiate printing with a single PRINT action. Online payment gateway integration is not included in Phase 1; payment, if collected, is handled manually by the shop (e.g., cash/UPI at the counter) outside the platform, and is proposed as Phase 2/Future Scope.

A key technical requirement is silent/direct printing without a normal browser print dialog. Because modern browsers do not permit unrestricted silent access to local printers, Phase 1 will use a secure local Windows Print Agent installed on the shop computer. The web application will create an authorized print job; the agent will retrieve the file securely and submit it to the configured Windows printer, then report the result to the backend.

Phase 1 is intentionally limited to document printing operations. Full stationery/product e-commerce is excluded and is proposed as Phase 2/Future Scope.

2. Project Objective

- Provide a single platform that can serve multiple independent printing shops.
- Give each shop its own account, QR code, pricing configuration, printer configuration and operational records.
- Allow customers to upload print-ready documents without requiring a complex customer registration process.
- Calculate printing charges from pages and selected printing options.
- Make print requests eligible for printing directly once options are confirmed; no online payment gateway integration in Phase 1.
- Enable one-click silent printing through a Windows Print Agent.
- Retain only the metadata needed for reporting after successful document deletion.
- Provide administrators with shop, pricing, print, QR, audit and system controls.

3. Business Workflow

Primary workflow:

1. Admin creates or activates a shop and configures its pricing and printer.
2. System generates a unique QR code for the shop.
3. Shop displays the QR code at its counter.
4. Customer scans the QR code and opens the shop-specific upload page.
5. Customer uploads a supported document.
6. System validates the file and calculates or determines page count where supported.
7. Customer selects printing options and reviews the final price.
8. Customer confirms the print request; the system marks the order eligible for printing (no online payment gateway step in Phase 1).
9. Shopkeeper sees the pending print job and clicks PRINT.

10. Backend queues the job for that shop's registered Print Agent.
11. Print Agent downloads the document through a short-lived signed URL and sends it to the configured Windows printer.
12. Agent reports success/failure; backend updates the job.
13. Only after confirmed successful printing does the retention/cleanup process become eligible to delete the document.

4. User Roles

Role	Key permissions
Platform Admin	Manage shops, users, pricing, QR codes, printers, transactions, print history, settings, reports, audit logs.
Shopkeeper / Shop User	Access only the assigned shop; view documents, print jobs, pricing/profile where permitted, printer status, history and summaries; initiate printing.
Customer	Scan QR, upload document, select print options, and view order/print status. No permanent account required by default.

5. Functional Requirements

5.1 Shop Management

- Shop fields: shop name, owner name, mobile, email, address, city, unique Shop ID, status and other required details.
- Admin can create, edit, activate and deactivate shops.
- Each shop has tenant-level data isolation.
- Shop credentials must be securely stored; passwords are hashed and never stored in plaintext.
- Deactivated shops cannot accept new customer print requests.

5.2 Customer / Document Flow

- Customer opens the shop-specific URL through QR.
- Supported baseline formats: PDF, JPG/JPEG and PNG. DOC/DOCX may be accepted only where a reliable server/agent conversion path is confirmed.
- Recommended maximum upload size: 25 MB per document, configurable.
- Document receives a unique document ID and is associated with shop_id.
- Customer sees filename, size, page count where available, selected options and total amount.
- Customer can track the order using a secure short-lived status token rather than creating a full account.

6. Admin Module

- Dashboard with shop, order and print summaries.
- Shop creation and activation/deactivation.
- User and role management.
- Pricing templates and shop-specific pricing.
- QR code generation, download, regeneration and deactivation.
- Printer/agent registration and status.
- Printing history and failed jobs.
- Reports and exports as agreed during UI finalization.
- System settings, retention settings and audit logs.

7. Shopkeeper Module

- Secure login.
- Incoming and pending print jobs.
- Document preview via authorized temporary URL.

- Print status.
- PRINT action for eligible jobs.
- Daily/monthly print summary.
- Shop profile and pricing visibility.
- Printer/Print Agent status where technically available.

8. Customer Module

- Scan QR and identify shop.
- Upload document.
- Select B&W/color, paper size and simplex/duplex where the configured printer supports them.
- View calculated amount.
- See print status.
- Receive a confirmation page. SMS/WhatsApp/email notifications are optional.

Recommended customer identity model: no mandatory registration for Phase 1. Each order receives a secure, unguessable reference/status token. If abuse, repeat customer accounts or refunds require stronger identity, an authenticated customer module can be introduced later.

9. Document Management & Lifecycle

Stage	Required behavior
Upload	Validate extension, MIME type, size and basic file structure; store in private S3.
Processing	Create metadata and calculate page count when supported.
Order confirmed	Document remains private and becomes eligible for print directly once options are confirmed (no online payment gateway step in Phase 1).
Print queued	Document remains available until confirmed success or controlled retry exhaustion.
Print success	Job marked successful; retention countdown starts.
Print failure	Document is retained for retry/investigation; never delete solely because a print command was attempted.
Cleanup	Scheduled job deletes eligible S3 objects and marks metadata as deleted while preserving required reporting metadata.

Recommended default retention: delete the actual document shortly after confirmed successful printing, for example within 15–60 minutes. Exact retention is a business setting and must be confirmed before production. A short window helps with operational recovery while limiting sensitive-document exposure.

10. Pricing Module

- Admin/shopkeeper can configure B&W per-page, color per-page and paper-size prices.
- Optional duplex/single-side pricing can be represented as a multiplier or separate price rule.
- Pricing must be versioned or snapshotted into the order so historical orders are not changed by later price edits.
- Total = applicable page count × applicable rate + any configured additional charge.
- Order stores the final calculated amount, currency, pricing rule/version and selected print options.

10.1 Page Count & Color Detection – Technical Note

PDF page count can be obtained reliably from the PDF structure. Color detection is more complex: a PDF page can contain text, images, vectors, transparency and embedded color spaces. Phase 1 will use a dedicated Python service (e.g., PyMuPDF) for server-side PDF analysis, invoked by the NestJS backend, to detect whether a page contains non-grayscale content, with a documented 'To Be Confirmed' accuracy threshold. For images, dimensions/color mode can be inspected. If exact billing-grade

color detection cannot be guaranteed for all PDF types, the customer/shopkeeper must be shown the detection rule and an admin override policy should be defined.

DOC/DOCX and other office formats should not be treated as guaranteed print-ready inputs unless a tested conversion service is included. PDF is the preferred production format for deterministic page counting and printing.

11. Payment Handling (Phase 1 - Out of Scope)

Online payment gateway integration is not included in Phase 1. Print requests become eligible for printing directly once the customer confirms the selected options and price - no online payment order, checkout, webhook or gateway verification step is part of this phase.

If a shop wishes to collect payment, it does so manually/offline (e.g., cash or UPI at the counter) outside the platform, and Phase 1 does not track or reconcile that payment.

Online payment gateway integration (e.g., Razorpay, PayU or Cashfree, subject to the client's merchant onboarding and account approval) is proposed as a Phase 2/Future Scope enhancement, at which point this section will define the gateway checkout, webhook verification, transaction recording and refund handling.

12. QR Code Module

Each shop receives a unique QR code. The QR should contain only a non-sensitive public route such as `https://print.example.com/s/{publicShopCode}`. It should not contain passwords or private document URLs.

- Admin creates shop → system generates a high-entropy public shop code.
- System creates a stable shop URL and QR image.
- Scanning opens the customer upload experience for that shop.
- Shop identity is resolved server-side; `order.shop_id` is assigned from the validated route.
- QR can be deactivated/regenerated without changing historical order ownership.
- QR images can be downloaded/printed by the admin.

13. Direct Printing Architecture

Silent printing is the most technically sensitive requirement. A normal browser cannot safely and consistently send a document directly to an arbitrary local printer without user interaction. Therefore Phase 1 will use a small Windows Print Agent/Service installed on each shop computer.

Architecture:

Customer Browser

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Angular Web Application — HTTPS —> NestJS API (Prisma + PostgreSQL)

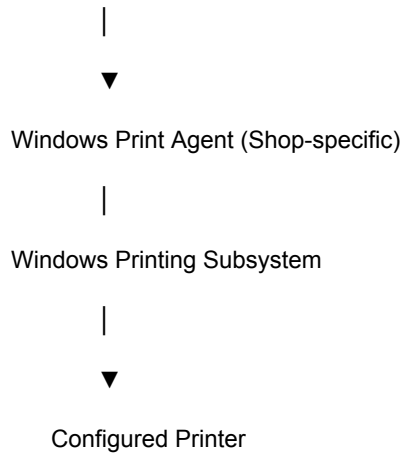
|



Print Queue (Redis + BullMQ) / DB

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WebSocket (preferred)



13.1 Print Agent Design

- Agent is installed on the shop's Windows computer and runs as a background service where feasible.
- Each agent receives a unique device/agent identity and credential; secrets are stored using OS-protected storage where practical.
- Agent is bound to one shop and cannot request jobs for another shop.
- Agent maintains a persistent secure connection to the backend using WSS/WebSocket. A polling fallback can be used if WebSocket connectivity is unavailable.
- Backend is the source of truth for job state; the agent is not trusted to change print-eligibility or order status. Job processing, retries and progress tracking are managed through a Redis-backed BullMQ queue.
- Agent receives only a job ID and authorized retrieval information; it downloads the document from a private S3 bucket using a short-lived signed URL.
- Agent submits the PDF/image to the configured Windows printer using a controlled print command/library.
- Agent reports accepted/printing/success/failure states with timestamps and an agent-side attempt ID.

13.2 Printer Configuration

- Each shop can have a registered printer and default printer setting.
- Phase 1 should target Windows-supported printers accessible through the shop PC's Windows print spooler.
- Printer name/driver must be tested during installation.
- Where the printer exposes supported paper size/color/duplex capabilities, the agent can validate requested options before submission.
- If the exact printer does not support a requested option, the job must fail safely or require a configured fallback rather than silently producing an unexpected output.

13.3 Failure Handling

Scenario	System response
Printer offline	Job remains pending/retryable; agent reports offline; no document deletion.
Paper unavailable / driver error	Agent reports failure; job remains retryable; shopkeeper sees error.
Internet unavailable	Agent cannot receive new cloud jobs; any locally queued job is handled according to the configured offline policy; backend remains authoritative.
Agent stopped	Backend marks agent heartbeat stale; shopkeeper/admin sees offline status.
Print command accepted but result uncertain	Job moves to PRINTING/UNKNOWN and requires controlled reconciliation; do not automatically delete.
Duplicate click	Backend idempotency/row locking prevents multiple active print attempts for the same job.

13.4 Recommended Print State Machine

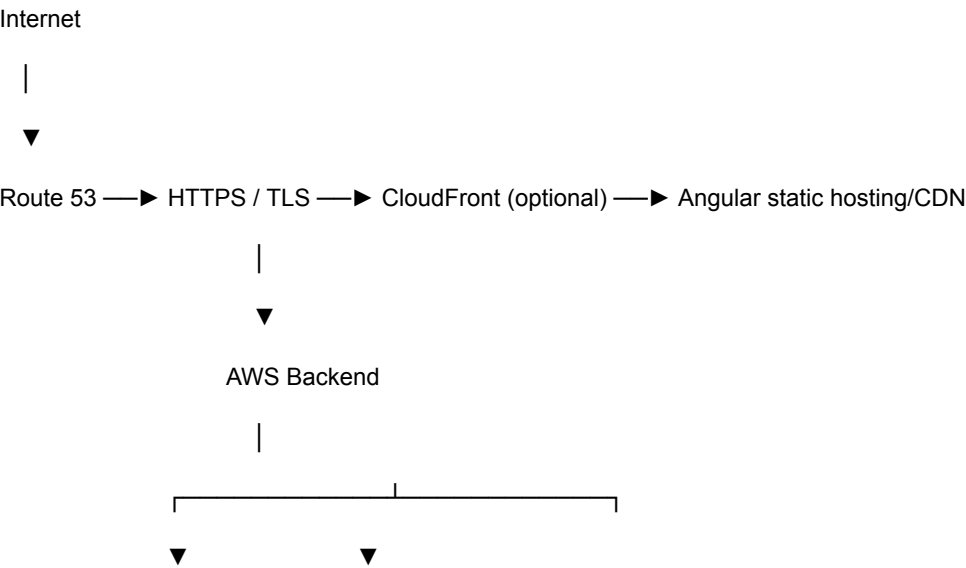
CREATED → PRINT_ELIGIBLE → QUEUED → PRINTING → PRINTED → RETENTION_PENDING → DELETED. Failure paths include PRINT_FAILED, AGENT_OFFLINE and PRINT_UNKNOWN. Transitions must be validated server-side.

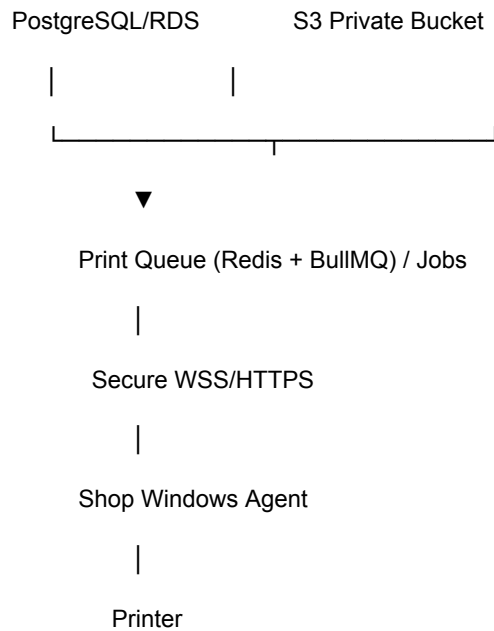
14. Technical Architecture

Recommended Phase 1 stack:

Layer	Technology	Purpose
Frontend	Angular + PrimeNG	Responsive customer, shopkeeper and admin web interfaces, built with PrimeNG components and PrimeFlex layout utilities.
Backend	NestJS (Node.js)	REST API, business logic and print orchestration (authentication delegated to Keycloak).
ORM	Prisma	Type-safe database access and migrations for PostgreSQL.
Auth	Keycloak	Centralized authentication/authorization (OAuth2/OIDC) for admin and shopkeeper logins.
Document analysis	Python service (e.g., PyMuPDF)	PDF page-count and color-detection processing, invoked by the backend.
Queue	Redis + BullMQ	Print-job processing, status tracking, retries and progress updates.
Database	PostgreSQL	Transactional data, tenant isolation, orders, settings and audit records.
File storage	AWS S3 private bucket	Temporary document storage with lifecycle/cleanup control.
Cloud	AWS	Production hosting and infrastructure.
QR	Server-side QR generation library	Unique shop QR generation.
Print	Windows Print Agent	Secure bridge from cloud platform to local Windows printer.
Monitoring	AWS CloudWatch	Application/system logs, alarms and operational visibility.

15. AWS Infrastructure





- EC2 is a practical Phase 1 backend option because it keeps initial infrastructure simple. A managed/containerized service can be adopted later.
- RDS PostgreSQL is recommended when budget permits because automated backups and managed database operations reduce maintenance risk. A PostgreSQL instance on EC2 is possible but increases operational responsibility.
- S3 bucket must be private; public object access is disabled.
- CloudFront can serve the Angular application and later front selected static content.
- Route 53 manages DNS where the client uses AWS DNS.
- TLS certificates should be managed through AWS Certificate Manager for supported endpoints.
- IAM roles should be used instead of long-lived AWS access keys on the server.
- Secrets should be stored in AWS Secrets Manager or Parameter Store.
- CloudWatch logs/alarms should cover backend health, errors, disk/memory, agent connectivity metrics where available.
- Backups: RDS automated backups and periodic recovery testing; S3 versioning/lifecycle policy as appropriate.
- Cost optimization: small production instance initially, S3 retention limits, log retention, scheduled non-production shutdown where applicable.

16. Database Design

All tenant-owned operational tables should carry `shop_id` directly or through a controlled foreign-key path. API authorization must enforce shop ownership on every read/write; filtering only in the frontend is not sufficient. Schema and migrations are managed through Prisma against PostgreSQL.

Table	Important columns	Keys / notes
shops	id UUID, shop_code, name, owner_name, mobile, email, address, city, status, created_at	PK id; unique shop_code
roles	id, name, description	PK id; predefined ADMIN/SHOPKEEPER
users	id UUID, shop_id, role_id, name, mobile, email, password_hash, status, last_login_at	PK id; FK shop_id/role_id; admin may have NULL shop_id
documents	id UUID, shop_id, original_name, s3_key, mime_type, size_bytes, page_count, color_pages, status, uploaded_at, deleted_at	PK id; FK shop_id
print_jobs	id UUID, shop_id, document_id, printer_id, options_json, amount, status,	PK id; FKs document/printer

	attempt_count, idempotency_key, queued_at, printed_at, failure_reason	
print_settings	id, shop_id, default_printer_id, retention_minutes, max_file_size	PK id; FK shop_id
pricing	id, shop_id, paper_size, color_mode, side_mode, price_per_page, effective_from, active	PK id; versioned price rules
qr_codes	id UUID, shop_id, public_code, target_path, status, generated_at, revoked_at	PK id; unique public_code
printers	id UUID, shop_id, agent_id, printer_name, driver_name, status, last_heartbeat_at, capabilities_json	PK id; FK shop_id
audit_logs	id UUID, actor_user_id, shop_id, action, entity_type, entity_id, ip, user_agent, metadata_json, created_at	Append-oriented; actor may be system
notifications	id UUID, shop_id, print_job_id, channel, event_type, destination, status, provider_ref, created_at	Optional notification subsystem
system_settings	key, value_json, updated_at	Global settings; restrict admin access

A payments table is not part of the Phase 1 schema, since online payment gateway integration is out of scope for this phase; it will be introduced in Phase 2 alongside gateway integration.

16.1 Relationships

- shops 1→N users
- shops 1→N documents
- shops 1→N print_jobs
- shops 1→N pricing records
- shops 1→N qr_codes
- shops 1→N printers
- documents 1→N print_jobs only if controlled reprint is allowed; otherwise enforce one active print lifecycle.
- print_jobs N→1 printer
- users N→1 roles

17. API Specification

Method	Endpoint	Purpose	Auth
POST	/api/auth/login	Admin/shopkeeper login	Public
POST	/api/auth/refresh	Refresh session/token	Authenticated
POST	/api/admin/shops	Create shop	Admin
GET	/api/admin/shops	List shops	Admin
PATCH	/api/admin/shops/:id/status	Activate/deactivate	Admin
GET	/api/shop/profile	Shop profile	Shopkeeper
POST	/api/documents	Upload document	Customer token / public signed session
GET	/api/documents/:id	Document details	Owner/status token
POST	/api/print/quote	Calculate print price	Customer/status token
POST	/api/print-jobs	Confirm print request (marks order print eligible; no payment gateway step)	Customer/status token
GET	/api/shop/print-jobs	Shop print queue	Shopkeeper
POST	/api/print-jobs/:id/print	Queue print job	Shopkeeper
POST	/api/agent/register	Register print agent	One-time provisioning/admin-controlled

GET	/api/agent/jobs/next	Polling fallback	Agent auth
POST	/api/agent/jobs/:id/status	Report print status	Agent auth
GET	/api/admin/qr/:shopId	Generate/retrieve QR	Admin
GET	/api/admin/print-history	Printing history	Admin
GET	/api/admin/reports/summary	Dashboard/report summary	Admin

17.1 Example – Price Quote

POST /api/print/quote

Request: { "documentId": "uuid", "paperSize": "A4", "colorMode": "BW", "sideMode": "SIMPLEX", "copies": 2 }

Response: { "quoteId": "uuid", "pageCount": 10, "billablePages": 20, "amount": 40.00, "currency": "INR", "expiresAt": "..." }

17.2 Example – Print Request

POST /api/print-jobs/{id}/print

Response: { "jobId": "uuid", "status": "QUEUED", "printerId": "uuid", "message": "Print job queued" }

17.3 Common errors

HTTP	Example	Meaning
400	INVALID_PRINT_OPTION	Unsupported option or invalid quote.
401	UNAUTHENTICATED	Missing/invalid authentication.
403	SHOP_ACCESS_DENIED	User attempted cross-shop access.
404	NOT_FOUND	Resource does not exist or is inaccessible.
409	JOB_ALREADY_PRINTING	Duplicate print attempt.
413	FILE_TOO_LARGE	Upload exceeds configured limit.
422	UNSUPPORTED_DOCUMENT	File cannot be processed safely.
503	PRINT_AGENT_OFFLINE	No connected agent for shop.

18. Security

- Keycloak-issued OAuth2/OIDC tokens for admin/shopkeeper APIs; use secure token storage and short access-token lifetime.
- Role-based access control and explicit shop-level authorization.
- Password storage and hashing handled by Keycloak's credential store; the application does not store passwords directly.
- HTTPS/TLS for all public and agent communications.
- Strict file extension + MIME + magic-byte validation; reject executable/script content.
- Configurable file size and upload rate limits.
- Private S3 bucket; no public object URLs.
- Short-lived signed URLs for document access.
- Rate limiting on login, upload, quote and public endpoints.
- Parameterized queries/ORM to prevent SQL injection.
- Output encoding and Angular security controls to mitigate XSS.
- CSRF protection for cookie-based sessions; if authorization is header-token based, use appropriate origin/CORS protections.
- CORS allowlist limited to the production application origins.
- Audit logs for administrative and sensitive operational actions.
- Document deletion and S3 lifecycle controls.
- Secrets stored outside source code.
- Malware scanning is recommended before production; exact antivirus/scanning service is To Be Confirmed and may add third-party cost.

19. Privacy & Data Protection

- Collect only information required to fulfill the print transaction.
- Keep uploaded documents private and temporary.
- Do not use customer documents for analytics, training or unrelated purposes.
- Restrict access to the relevant shop and authorized operational users.
- Use encryption in transit and AWS S3 server-side encryption at rest.
- Retain only necessary order metadata after document deletion.
- Define a documented retention policy before production.
- Maintain an appropriate privacy policy and customer notice for document handling.
- Backups should be treated as a separate retention consideration; if documents are ever included in backups, the backup lifecycle must also be governed by the retention policy.

20. Notification

Phase 1 will provide in-application status as the baseline. Optional channels can include email, SMS or WhatsApp for upload received, print queued, print completed and print failed. Provider fees and template/registration requirements are third-party costs and are excluded unless explicitly added.

21. Development Plan – 4 Weeks

Week	Activities	Deliverables
Week 1	Requirement finalization; UI/UX; DB schema; project setup; environments; authentication; roles	Approved flows, initial UI, DB migrations, login/RBAC foundation
Week 2	Admin/shop modules; customer upload; S3; document metadata; pricing/quote	Shop/admin dashboards, upload flow, S3 integration, pricing engine
Week 3	QR; Print Agent; printer registration; direct print workflow	Shop QR, Windows agent prototype and integrated print flow
Week 4	End-to-end testing; failure scenarios; security review; deployment; documentation; demo; UAT; launch	Production release candidate, test evidence, documentation, deployment and handover

Timeline dependency: the 4-week target assumes prompt client decisions, availability of at least one representative Windows PC and printer for testing, AWS/domain access, and timely UAT feedback. Printer-driver incompatibility or major scope changes can extend the schedule.

22. Testing Strategy

- Unit tests for pricing, authentication, authorization, state transitions and utility functions.
- API tests for authentication, shop isolation, uploads and print endpoints.
- Integration tests for S3, database and Print Agent.
- UI tests for customer, shopkeeper and admin critical journeys.
- Print testing: PDF, image, color/B&W, simplex/duplex where supported, multiple copies.
- Printer failure testing: offline, paper/driver error, agent disconnected, duplicate click.
- Network failure testing between cloud and agent.
- Security testing: authorization bypass, file upload abuse, rate limits, XSS, SQL injection patterns and secret exposure.
- Cross-browser testing for current Chrome/Edge and a supported mobile browser.
- AWS deployment, backup and recovery verification.
- User Acceptance Testing (UAT) with agreed acceptance checklist.

23. Acceptance Checklist

- Admin can create/activate/deactivate at least 10 shops.

- Each shop has isolated login and cannot access another shop's documents/orders.
- Each shop can have a unique QR code that resolves to the correct shop.
- Customer can upload a supported PDF/image file.
- Customer can select configured printing options and receive a calculated amount.
- Confirming the print request creates an eligible print job without any online payment gateway step.
- Shopkeeper sees the pending print job and can click PRINT.
- Representative Windows Print Agent receives the job and prints to the configured test printer without the normal browser print dialog.
- Agent reports successful/failed status.
- Failed printing does not delete the document.
- Successful printing triggers the configured retention/cleanup process.
- Admin can see print history.
- Basic audit records are available for sensitive actions.
- Production HTTPS, private S3 and configured backups/monitoring are operational.

24. Deployment Plan

14. Development environment: local Angular, NestJS + Prisma, PostgreSQL, a local Keycloak instance, the Python PDF-analysis service, and a development S3 configuration.
15. Staging: AWS resources mirroring production at a smaller scale; representative printer/agent.
16. Production: domain, HTTPS, backend, database, private S3, monitoring, secrets and scheduled cleanup.
17. Run database migrations with version control.
18. Configure environment variables/secrets through a secure secrets mechanism.
19. Configure RDS backups and test a recovery procedure.
20. Deploy backend and frontend using a repeatable release process.
21. Install and register Print Agent on the pilot shop computer.
22. Complete UAT and obtain client sign-off.
23. Maintain rollback capability using the previous backend build and database migration strategy.

25. Maintenance & Support – 1 Year

One year of maintenance is included subject to the commercial agreement. Maintenance begins from production handover.

Included

- Bug fixes for functionality delivered in Phase 1.
- Security updates to supported application dependencies when reasonably required.
- Minor technical support related to the delivered system.
- Deployment assistance for bug-fix releases.
- Basic server/application monitoring support.
- Backup monitoring and guidance.
- Troubleshooting of the delivered Print Agent software where the supported Windows/printer environment remains within the agreed compatibility baseline.

Not Included

- New modules or business workflows.
- Major UI redesign.
- New payment, messaging or external integrations.
- Major business-rule changes.
- Phase 2 e-commerce/stationery features.
- New printer hardware or driver procurement.
- Third-party subscription or transaction charges.
- Major AWS architecture changes or scaling projects.
- On-site support unless separately agreed.
- Custom support for printers/Windows environments outside the agreed test matrix.

New requirements are treated as Change Requests and separately estimated and billed.

26. Scope & Exclusions

Phase 1 Included

- Multi-shop management up to the agreed initial shop count (commercial assumption: up to 10 shops for initial deployment).
- Customer document upload and print ordering.
- Pricing and quote calculation.
- Shop-specific QR codes.
- Admin and shopkeeper dashboards.
- Windows Print Agent and silent direct printing for the agreed test printer/Windows environment.
- Document retention/deletion workflow.
- Reports and audit logs at the level specified in this document.
- Production deployment and one-year maintenance as defined.

Explicitly Excluded from Phase 1

- Online payment gateway integration (e.g., Razorpay/PayU/Cashfree); payment, if collected, is handled manually/offline by the shop.
- Full e-commerce/stationery marketplace.
- Stationery product ordering.
- Book ordering.
- Stationery purchasing.
- Inventory/stock management.
- Delivery management.
- Advanced loyalty system.
- Advanced analytics beyond agreed operational summaries.
- Native Android/iOS customer applications.
- Multi-city operations beyond the initial architecture/configuration unless separately agreed.

27. Future Phase 2

- Online payment gateway integration (e.g., Razorpay/PayU/Cashfree) with verified webhooks and transaction records.
- Stationery product catalog and online store.
- Books and stationery ordering.
- Inventory management.
- Delivery and delivery-partner workflows.
- Offers and coupons.
- Wallet and loyalty points.
- WhatsApp/SMS automation.
- Advanced analytics and business intelligence.
- Customer accounts and repeat-order profiles.
- Multi-city expansion and franchise-level controls.

28. Project Risks & Mitigation

Risk	Impact	Mitigation
Browser silent-print limitation	High	Use dedicated Windows Print Agent; do not depend on browser silent printing.
Printer/driver compatibility	High	Define supported Windows/printer baseline and test representative devices before launch.
Scanner compatibility	Medium	Customer upload is device-agnostic; scanning is performed by user's

		existing device/scanner. No scanner-driver integration in Phase 1.
Internet outage	High	Agent heartbeat/status, retryable queue and clear offline status; backend remains source of truth.
Large/malicious files	High	Size limits, type validation, private S3, optional malware scanning.
Duplicate print	High	Server-side state machine, locking and idempotency keys.
Failed printing	High	Do not delete until confirmed success; retry/reconciliation states.
AWS cost growth	Medium	Retention controls, small initial resources, monitoring and budget alerts.
Privacy exposure	High	Private storage, least privilege, short retention and audit logs.
Color detection ambiguity	Medium	Define tested detection rule; show billing logic; admin override/exception process if required.
Scope expansion	High	Formal Change Request policy and Phase 2 separation.

30. Third-Party Costs

- Domain registration/renewal.
- AWS EC2/RDS/S3/CloudFront/Route 53/CloudWatch and related infrastructure charges.
- SMS/WhatsApp/email provider charges, if used.
- Paid SSL certificate, if the chosen hosting setup requires one; standard AWS-managed TLS may avoid a separate certificate fee.
- Printer/scanner hardware, consumables and replacement parts.
- Windows licensing or required third-party software.
- Malware scanning or document conversion services, if selected.

These costs are separate from the platform's development scope and are payable by the client unless a separate written agreement explicitly says otherwise.

31. Client Responsibilities

- Provide final business rules, shop details and authorized contacts.
- Provide domain/DNS access or authorize required configuration.
- Provide AWS account/billing ownership or approve the infrastructure arrangement.
- Provide at least one representative Windows computer and printer for direct-print testing.
- Provide timely feedback and UAT sign-off.
- Provide final pricing rules and print-option definitions.
- Provide legal/privacy/terms content where required.
- Pay third-party operational costs directly or reimburse them as agreed.

32. Developer Responsibilities

- Design and develop the Phase 1 application according to the approved SRS.
- Implement backend, frontend, database, S3 integration and QR workflow.
- Develop the Windows Print Agent for the agreed Windows/printer test environment.
- Implement authentication, shop-level authorization, audit logging and document retention.
- Perform agreed testing and fix Phase 1 defects.
- Deploy the system to the agreed AWS environment.

- Provide technical documentation and basic handover.
- Provide one-year maintenance within the defined support scope.

33. Change Request Policy

Any requirement not explicitly included in the approved Phase 1 scope is a Change Request (CR). Examples include new integrations, additional workflows, major UI redesign, additional printer ecosystems, product ordering, inventory, delivery, loyalty or advanced analytics.

24. Client submits the requested change.
25. Developer assesses impact on effort, timeline and dependencies.
26. Developer provides a written estimate.
27. Client approves the CR and any additional fee before development starts.
28. Approved CR is scheduled without compromising committed work unless both parties agree otherwise.

34. Acceptance Criteria & Warranty

The project is considered accepted when the Phase 1 acceptance checklist is substantially satisfied in the agreed staging/production environment and the client confirms UAT completion. Defects in delivered functionality reported during the agreed maintenance period are handled under the maintenance terms. Third-party outages, unsupported hardware, client-side infrastructure changes and new requirements are not considered defects in the delivered software.

35. Final Conclusion

Phase 1 provides a practical foundation for a scalable multi-shop printing platform: each shop receives isolated access and a unique QR code; customers can upload documents and request prints; shopkeepers can manage print requests; and a dedicated Windows Print Agent enables the required silent/direct printing workflow. Online payment gateway integration is not part of Phase 1; any payment collection is handled manually by the shop, outside the platform. The architecture deliberately separates temporary document storage from durable metadata, reducing privacy risk and storage cost.

One year of defined maintenance is included. AWS, domain, messaging and hardware costs remain separate. Online payment gateway integration and advanced stationery e-commerce functionality are intentionally deferred to Phase 2.

Scope Summary

Area	Phase 1 commitment
Shops	Multi-shop; initial commercial assumption up to 10 shops
Customer	QR → upload → options → confirm → status
Documents	PDF/JPG/JPEG/PNG baseline; other formats subject to technical confirmation
Pricing	Configurable page/print-option pricing
Payment	Not included in Phase 1 (offline shop collection if needed); online gateway deferred to Phase 2
Printing	Silent/direct printing via Windows Print Agent
Storage	Private AWS S3 with controlled retention/deletion
Admin	Shops, users, pricing, QR, prints, reports, settings, audit
Shopkeeper	Queue, preview, print, status, history, summaries, printer status
Timeline	Approximately 4 weeks, subject to dependencies
Maintenance	1 year as defined
Phase 2	Stationery/product e-commerce and advanced features excluded

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